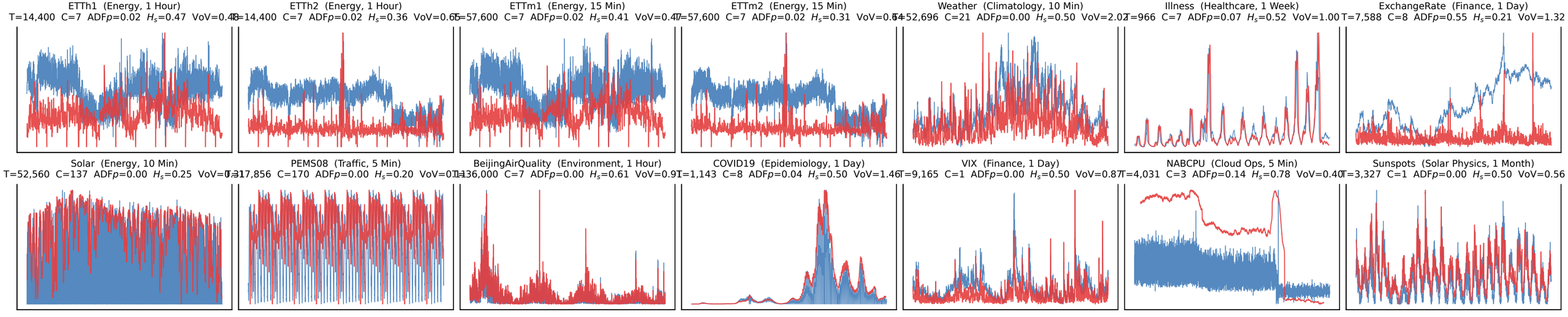Local heterogeneity across benchmark datasets: signal (blue) and rolling std (red) reveal regime shifts and volatility-of-volatility